

Specifications of individual components of NOVAC scanner systems

1. High latitude scanner, cone geometry

These are DOAS scanners made of plastic. They can be disassembled into a scanner mounting component, a motor mounting component and a protective hood. The hood covers the optical mirror and scanner motor. They are delivered with four custom-made bolts that connect the individual components of the scanner. Two holes are provided for mounting two fiber optics cables – thus allowing dual beam measurements. The high latitude version of the scanners are modified for use in high-latitude environments. In contrast to the standard scanners, these use a fused-silica tube through which the scanning mirror or prism collects scattered UV light. This eliminates the need for external moving parts which would otherwise be susceptible to rime ice. The fused silica tube is heated by excess heat from the stepper motor.

2. Metal pole holder for mounting scanner on mast

This component is used to mount the scanner on a steel pipe frame or mast. One side fits within the scanner mounting component. Bolts are provided to attach the two. The other side of the pole holder can be clamped or bolted to a steel frame or mast.

3. Motor cable, 6 m long, with DB9 adapter

These cables are used to connect the scanner stepper motor with the NOVAC instrument typically installed at the base of the scanner mast. The motor cables have special connectors that allow them to be passed through narrow openings and conduit within the scanner and instrument enclosures. The short adapter breaks out one end of the cable to a DB9 connector for attachment to the data acquisition system.

4. Cylindrical quartz window

These cylindrical lenses allow the updated scanners to collect scattered UV radiation from a wider range of angles than previously possible. The new design is advantageous for wind speed measurements using the dual beam capabilities of the scanner systems.